Warm Up:

What are some primitive types in Python?

Particle Motion and Kinematics

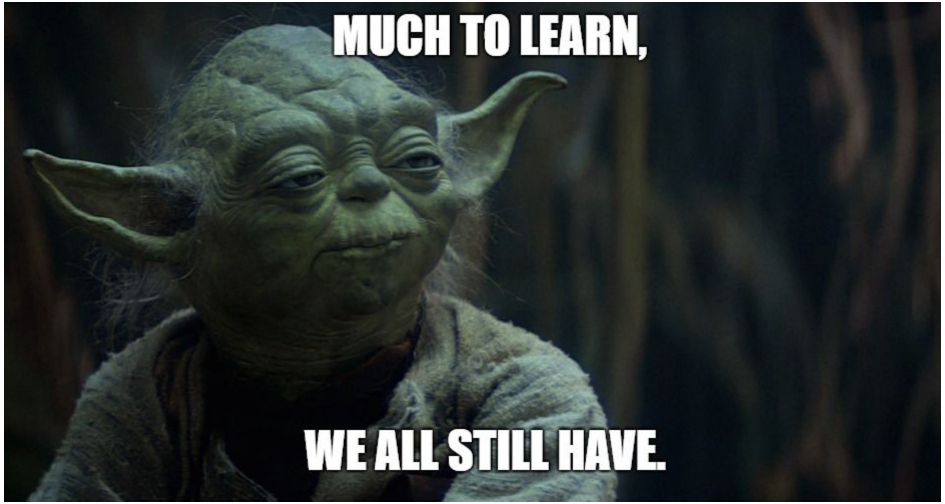
SciTeens SOLID Physics Day 2

Recap

- Last lecture, we discussed the following:
    - Vectors and their relevance in particle physics
    - Specific vectors in particle motion (i.e. position, velocity, and acceleration)
    - The relationships between position, velocity, and acceleration with respect to time
- Today we will be expanding those concepts into kinematics and 1-D particle physics

Learning Objectives

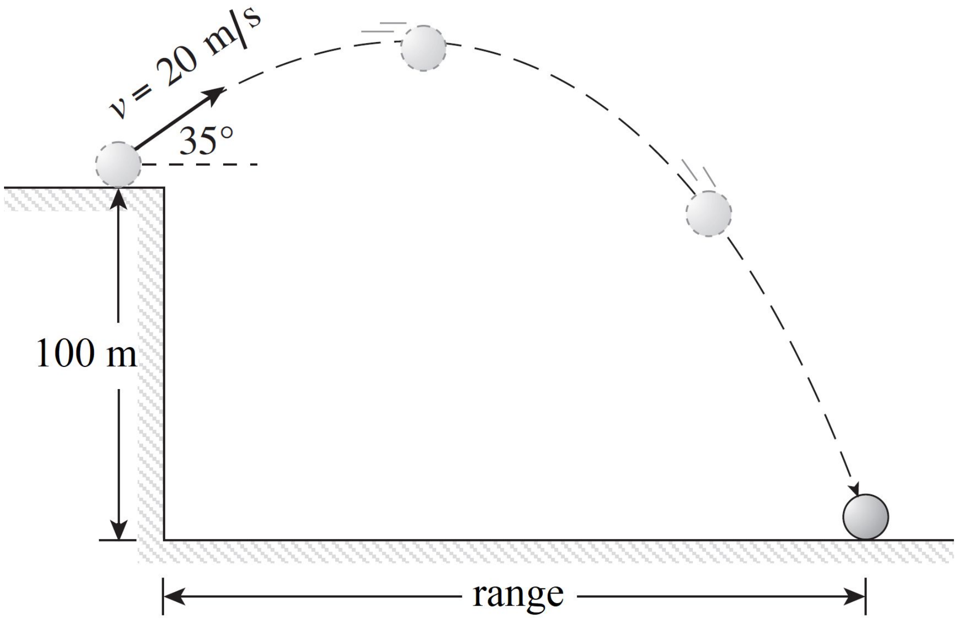
- Kinematics
  - What are they?
  -
- The 4 kinematic equations
  - How to use them and when
  - Examples of when to use them
- Practice
  - 1 dimensional kinematics
  - Word problems with kinematics



What is Kinematics?

Kinematics is the study of projectile motion without reference to the forces that result in the motion itself

- It allows for a simple yet versatile view of particle motion
- Forces ignored include friction, air resistance, centripetal forces, etc.



What Can Kinematics Represent

The kinematic equations can represent any particle in motion in as many dimensions as possible as long as the following conditions are satisfied:

- The particle has a constant acceleration ($\Delta a = 0$)

That's it! Therefore, the equations are extremely flexible and can be used for almost any scenario you can imagine.

The Kinematic Equations

Remember the relationships between projectile motion vectors that we discussed? The kinematic equations translate those relationships into mathematical formulas!

1. $v = v_0 + at$
2. $\Delta x = (\frac{v + v_0}{2})t$
3. $\Delta x = v_0 t + \frac{1}{2}at^2$
4. $v^2 = v_0^2 + 2a\Delta x$

Checkpoint 1

Of the following, which is NOT a variable found in the 3rd kinematic equation (the most important one?)

(a) v (final velocity)
(b) Δx (change in position)
(c) a (acceleration)
(d) t (time)

Kinematic Equation #1

$$1. \quad v = v_0 + at$$

This equation is an extension of defining acceleration as a function of velocity divided by time; algebraically then, velocity can be defined as a function of the product of acceleration and time

- v is the final velocity, and v0 is the initial velocity
- a is the acceleration over the time period, and t is the time period itself

Kinematic Equation #2

$$2. \quad \Delta x = \left(\frac{v + v_0}{2}\right)t$$

This equation involves a clever use of algebra assuming acceleration is constant

- v is the final velocity, and v0 is the initial velocity, so the term $\left(\frac{v + v_0}{2}\right)$ is the “average velocity” over the time period
- Δx is the change in position (or displacement), can be defined using the equation Δx = x - x0 where x is the final position and x0 is the initial position

Kinematic Equation #3

$$3. \quad \Delta x = v_0 t + \frac{1}{2} a t^2$$

This equation is the one most commonly used to determine the position of a particle, and the values used are often provided.

- Δx is the change in position, adhering to the same definition as before
- v0 is the initial velocity
- a is the acceleration over the time period, assuming acceleration remains constant (we assume this throughout the course)
- t is the time period, note that it appears twice in the equation

Kinematic Equation #4

$$4. \quad v^2 = v_0^2 + 2a\Delta x$$

This equation is the least used but still important; note that this is the only kinematic equation in which time (t) is absent

- v is the final velocity, and v0 is the initial velocity
- a is the acceleration over the time period, assumed to be constant
- Δx is the change in position over the time period, which is unknown in this case

Differentiating Between Equations

Use equation #1 when:

- x is irrelevant and the focus is on v and v_0

Use equation #2 when:

- a is unknown but the focus is still on net displacement

Use equation #3 when:

- net displacement is desired and v_0 and a are given

Use equation #4 when:

- t is unknown and the problem deals with the change in velocity

Examples: Differentiating Between Equations

Ex 1: Given v_0 = 10, a = 2, and t = 10, what kinematic equation is most appropriate to finding v? Solve the equation for v.

Ex 1: Given v_0 = 5, Δx = 100, and t = 5, what kinematic equation is most appropriate to finding a? Solve the equation for a.

Ex 1: Given v = 10, Δx = 100, and v_0 = 5, what kinematic equation is most appropriate to finding a? Solve the equation for a.

Checkpoint 2: Example Problem

Suppose a particle can move horizontally either East or West, where East is the +x direction. The particle's initial velocity is 1m/s, its acceleration is 5m/s/s, and it travels for 10s before instantaneously stopping. Assume that the particle starts at $x0 = 0m$.

(1) What is the displacement of the object from its initial position. Is this value the same as its final position?

(2) What is the velocity of the particle the moment right before it stops? (Essentially at $t = 10$)

(3) If we assume that the particle starts at $x0 = -10m$, which of the following values change: $v0$, v, x, Δx?

Practice Problems

Because practice makes perfect!

[Kinematics Practice #1]

[Kinematics Practice #2]

[Kinematics Practice #3]

Additional Practice/Resources

Resource #1

Resource #2

Resource #3

Additional Practice #1

Additional Practice #2